EnsembleSE-Indic: Diverse-Member Neural Ensembles for Noise Event Removal in Indian-Language Speech

IndoML Datathon 2026, Track 2 — Team marchmello01

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***Abstract*—Track 2 asks for the removal of transient noise events from Indian-language speech, scored by SI-SDR on synthetic mixtures plus the reduction in word error rate of the mandated SraVaani recogniser on all clips. We submit a single audio-only pipeline applied identically to every clip: a fixed ensemble of four enhancement architectures (MP-SENet, URGENT BSRNN, TF-GridNet and TF-Locoformer; seven checkpoints) with eight-view test-time augmentation. All networks are fine-tuned with an SI-SDR loss plus a recognition (RNNT) loss through the frozen SraVaani model, on mixtures of released and leakage-filtered public Vaani speech with reused validation noise events. The largest late gains came from averaging diverse TF-Locoformer members rather than training one model longer. The final system scores 19.8688 Combined on the Phase 2 test set (SI-SDR 16.29 dB, pooled ΔWER 3.58); our Phase 1 system placed 2nd.**

I. Training Data and Preprocessing

Released data. We use the released Track 2 training clean/noisy pairs, holding out two Indian states for validation, the released natural clips (only for a preservation loss), and the released validation set.

Reused validation noise ("d32" bank). From the released validation pairs we extract 518 noise-event recordings as noisy-minus-clean residuals. The organizers confirmed that training on validation noise is allowed. The bank is used for training only; inference uses no noise recordings.

Public Vaani speech (additional data). We add 6,069 clean utterances from the public Vaani corpus [1]. To prevent test leakage we build an audio landmark-fingerprint index of all 3,178 Phase 2 test inputs, drop every Vaani recording matching a test input with ≥ 12 votes, drop all recordings of any speaker with ≥ 20 matches, and exclude the two held-out states. Targets are pause-gated to match the convention of the organizers' clean references. The organizers confirmed that public Vaani training is allowed.

Augmentation. Every trainer remixes on the fly: clean speech plus d32 events (placed at +9 ± 3 dB event-to-speech level until about 34% of the clip is covered), released training residuals, or the clip's own noise, followed by peak normalisation to 0.95.

II. Proposed Method

Architectures and pretrained models. (i) MP-SENet [2], magnitude/phase enhancement, from the public DNS checkpoint; (ii) BSRNN from the URGENT challenge baseline [3]; (iii) TF-GridNet [4], from the URGENT baseline; (iv) TF-Locoformer [5], from MERL's public DNS checkpoint. SraVaani-1.0 [6] (ARTPARK-IISc, Hugging Face) is used frozen for the recognition loss and for transcription; ESPnet [7] supplies the STFT front end.

Loss. On synthetic mixtures

$$\mathscr{L} = 2 \cdot (-\text{SI-SDR}(\hat{s}, s)) + 3 \cdot \mathscr{L}_{RNNT}(\hat{s}, y),$$

where $\mathscr{L}_{\text{RNNT}}$ passes the enhanced waveform through SraVaani's frozen encoder and joint network with the released transcript y. On natural clips (no clean reference) we use $0.2 \cdot \|\hat{s} - \tilde{s}\|_1 / \|\tilde{s}\|_1$ against a frozen teacher output $\tilde{s}$, which prevents the enhancer from damaging real recordings. Optimiser: AdamW, gradient clipping at 3.

Training strategy. Each network is first adapted on released data, then on d32 remixes and on the leakage-filtered Vaani speech. Continuation rounds of 40k steps at learning rate $5 \cdot 10^{-6}$ (single H100/H200 GPU each) refined the TF-Locoformer and TF-GridNet. The final TF-Locoformer slot averages four checkpoints from different lineages: the scaled Vaani recipe (60k steps plus four continuation rounds), a released-data recipe with natural preservation continued to 33k steps, and two reused-noise specialists (d32-only mixtures) initialised from different points of the first lineage.

Inference. The ensemble is fixed: b = 0.3·MP + 0.7·g·BSRNN (g: least-squares gain between the two outputs), c = 0.3·b + 0.7·TFGridNet, and ŝ = 0.3·c + 0.7·Loc$\bar{o}$, where Loc$\bar{o}$ is the equal average of the four peak-normalised TF-Locoformer outputs. BSRNN input is level-normalised to RMS 0.10. Each network runs on eight views (polarity ±1 × shifts of 0, 40, 80, 120 samples), which are inverted and averaged. Every clip receives identical processing: no routing, passthrough, noise dictionary or template search.

Phase 1 system. Our Phase 1 entry (2nd place) used a binary CRNN noise-event span detector and two event-local MP-SENet experts fine-tuned with an ASR loss; the code and checkpoints are in the repository's *phase1/* folder.

III. Results and Conclusion

Table I. Official Phase 2 results of successive systems

System	SI-SDR	ΔWER syn	ΔWER nat	Combined
Ensemble, 1 Loco (F6)	16.048	10.174	0.353	19.547
+ continuation rounds (F11)	16.170	10.263	0.370	19.709
+ round 3, TTA×8 (F12b)	16.197	10.299	0.379	19.754
+ 3 diverse Loco (F13)	16.270	10.432	0.357	19.854
+ 4th Loco member (final)	**16.289**	**10.432**	**0.350**	**19.869**
3 Loco, longer training (F14)	16.280	10.374	0.341	19.835
+ 5th Loco member (F16)	16.286	10.452	0.328	19.857

Table I lists official Phase 2 scores (Combined = SI-SDR + pooled ΔWER; pooled ΔWER ≈ $0.35 \cdot \Delta\text{WER}_{\text{syn}} + 0.65 \cdot \Delta\text{WER}_{\text{nat}}$). The final system reaches **19.8688**.

Key findings. (1) *Diversity beats depth.* Adding differently-trained TF-Locoformer members gave the largest late gain (+0.100 for F13), while continuing to train existing members gave smaller, eventually negative returns (F14). (2) *Only SI-backed gains transfer.* Candidates whose local gain was purely a few fewer ASR errors (F14, F16) scored lower officially; candidates that also improved hard-clip SI-SDR transferred. (3) *Recognition-specific objectives did not transfer.* A matched TDT loss, RNNT weights of 10× and 25×, an artifact-boosted SDR loss, teacher labels, a generative Schrödinger-bridge enhancer and second-stage denoisers were all neutral or negative under the exact package. (4) *Where the remaining error is.* On held-out synthetic clips our output leaves 615 SraVaani word errors against 507 for the clean reference; 71% of that gap is substitutions (residual distortion), not deletions. (5) *Evaluate the shipped package.* A cascade that looked +0.45 pooled-ΔWER better in a no-TTA harness gave nothing once evaluated package-to-package with test-time augmentation.

Conclusion. A compliant, uniform neural ensemble reached 19.87 Combined in Phase 2, with the 2nd-best SI-SDR among the top teams, after a 2nd-place Phase 1. Its main limitation is synthetic ΔWER: the recogniser still substitutes words on our enhanced speech. Closing that gap appears to need enhancement objectives that track recognition more directly than the surrogate losses we tested. Code and checkpoints: github.com/marchmello1/EnsembleSE-Indic.

References

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